

# Improving Platform Interoperability for Efficient Patient Care

In today's radiotherapy environment, quick and easy access to patient data is critical to providing the best possible patient care and ensuring facilities operate at maximum efficiency. Many oncology clinics choose the ARIA oncology information system (OIS) for a best-in-class approach to cancer care. They, along with the other clinics in their integrated healthcare system, use a different platform for their hospital-based electronic medical records (EMR).

Without integration between ARIA® and other systems, however, clinics experience the challenges of two separate systems, including duplication of effort, manual workarounds, missing documentation, and the potential for medical error. For TriHealth, an integrated health system based in Cincinnati, Ohio, lack of interoperability between ARIA and Epic, their hospital EMR, meant duplicative processes, inefficient billing practices, and low throughput. When Michael Grindstaff, Director of Radiation Oncology joined the organization, he set out to improve its clinical efficiency, implement best-practice standards and protocols, and make use of powerful analytics.

**...each care team was doing things differently and having to enter data in multiple systems...**

"Without standardized care, processes and protocols, our throughputs were suffering, and we couldn't take advantage of data to give us meaningful insights into opportunities to grow," Grindstaff explains. "Furthermore, each care team was doing things differently. Having to enter data in multiple systems was slowing everyone down."



## Designing a Comprehensive Workflow

To help data flow seamlessly across the ARIA and EMR systems, consultants from Varian Professional Services works with the EMR company and their mutual customers to develop a comprehensive solution that helps ensure continuity, efficiency, and safety. By leveraging ARIA Connect, Varian consultants manage the necessary HL7 interfaces, configuration, and other services required to help ensure the predetermined level of ARIA and EMR software integration across clinics and hospitals.

Varian consultants combine deep clinical knowledge of oncology workflows with OIS expertise. They leverage their proprietary interfaces to produce custom integration plans that meet the needs of each hospital and clinic. Varian consultants confer with key stakeholders and take the clinics through each essential step of interoperability, helping ensure clinicians have access to the data they need when they need it.

In the case of TriHealth, the Varian Professional Services team worked with TriHealth as part of a major initiative to integrate radiation oncology practices across its multiple facilities onto one common software platform and database.



"As TriHealth's new director, I sat down with a team of clinical experts within our department to review our gamut of software and hardware systems and the few interfaces already in place." Grindstaff says. "Ultimately we decided on Varian Professional Services for help. Varian put together the right team to help us see the big picture and how everything needed to come together."

**Varian put together the right team to help us see the big picture and how everything needed to come together.**

### Data Integrity Across the Care Continuum

The TriHealth project involved reconfiguring patient data and treatment systems to standardize clinical delivery workflows across the enterprise network, which comprised three operational sites, and two that were in the process of coming online. Full data exchange was required across several Varian and third-party machines and software applications.

TriHealth's goal was to standardize care across regional facilities and improve efficiency by creating a sustainable, patient-centered, end-to-end integrated service. To do this, Varian Professional Services worked with Grindstaff and his team to ensure data integrity for the care continuum within the complex radiation oncology treatment framework and the hospital system.

### Customizable to Clinical Needs

Varian Professional Services offers a wide range of consulting services based on best practices and clinical expertise and can design a customized plan for full software integration, depending on the size and needs of the clinic. Resources and processes may vary depending on the scale of the project, but typical steps include:

- Clinical observation and assessment of current workflows and software use
- Project kick-off, including project governance
- Design phase to confirm workflows and standardize terminology
- Interface builds
- Interface QA testing
- Go live
- Monitor and support maintenance as required

### Rapid Time to Deployment for TriHealth

The TriHealth project involved each of the steps outlined above and involved a concentrated team of Varian consultants. The first implementation went live in about six months, and according to a joint analysis by Varian and TriHealth, achieved an accelerated time saving of about a third compared to similarly sized projects. The effort required from TriHealth included representation from nursing staff, physicists, dosimetrists, radiation therapists, billing, and physicians who were involved as well as the operations committee administration for about an hour a week each.

"Not only did we ensure we had executive support from the top down, but we formed a project committee early and ensured we had the right people—from reception to physicians and nurses," Grindstaff says. "Varian explained the rationale behind workflows and helped with decision points. Everyone from Varian Professional Services put tireless hours into making sure things were right."

### Data in the Right Place at the Right Time

As each center went live, TriHealth saw immediate benefits from system integration. Consult notes, pathology, microbiology and surgical records created in the EMR system automatically interfaced into ARIA, OTV, CTPN (critical treatment planning notes), CT Sim notes, EOT summaries, and RO notes completed in ARIA were shared in the EMR platform. According to TriHealth's data, this translated to a 25% increase in patient data shared between the two systems.

"No longer having to repeat the amount of documentation in two different systems has incurred massive savings," Grindstaff says. "Initially, information crossover was very light. Now we have a 1:1 relationship and the right documents are in the right place for the right people at the right time."

### Streamlined for Safety and Efficiency

Improving efficient workflows and interoperability between ARIA and EMR platforms is helping many clinics like TriHealth streamline operations while reducing the opportunity for clinical and financial errors. Easy access to consistent information gives clinicians confidence in their data and patients see timely, accurate portal information. Other benefits from a Varian Professional Services engagement for integration include:



#### Clinical Documentation

All documents are available when and where they are needed so clinicians can access the full patient chart, including details on other health conditions. Physician documentation relevant to a procedure can easily be verified to meet United States regulatory requirements to support reimbursement.



#### Billing

Each episode account number from the EMR platform is automatically connected with each episode in ARIA, so financial numbers are consistent, and documents are filed, sorted, and populated correctly. This supports accurate and complete billing.



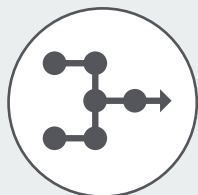
#### Scheduling

With drag and drop ease, all required activities, such as weekly on-treatment visits and chart checks, are populated across ARIA and the EMR platform. If there is a modification in an appointment scheduled in ARIA, the EMR platform is automatically modified, too.



#### Reporting and Data Analytics

Improved efficiencies through streamlined analytics result in rich operational reporting, including weekly activity metrics, patient referral statistics, diagnoses, and insurance. Clinics have increased confidence in data integrity and security.



#### Workflows

Varian Professional Services works with clinicians to review workflows and flowcharts by discipline to identify opportunities to streamline operations. This can inform the planning process for care coordination across multiple clinics and helps improve throughput.



#### Best Practice Protocols

Best practices for clinical protocols can be fully populated to help ensure adherence and provide standardized care across care teams.

## A Platform for Growth

As the largest healthcare provider in the Cincinnati region, TriHealth serves approximately one in three patients at any given time and continues to expand. Grindstaff acknowledges that the benefits of interoperability support this growth, and he's already looking at ways to use reporting data and analytics to improve the patient experience, standardize treatments, and manage resources across clinics.

"We now have an excellent understanding of what's going on with patient care outside of radiation oncology," Grindstaff explains. "For instance, there's a lot less double booking and rescheduling because schedulers can see the patient's full schedule, access more details on specific procedures, adjust times, and help patients better prepare for visits."

For more information on the levels of engagement Varian Professional Services can provide to help your facility pave the way to integrated patient data between ARIA and other software platforms, including the Varian Executive Partnership Program, contact your Sales Representative.



**...we now have an excellent understanding of what's going on with patient care outside of radiation oncology.**

### Varian Professional Services and TriHealth At-A-Glance

About TriHealth: Based in Cincinnati, Ohio, TriHealth comprises five hospitals with more than 11,500 staff. The TriHealth Cancer Institute uses a mix of radiation equipment and hardware, including ARIA, Eclipse™, TrueBeam® and other third-party systems across its facilities.

Goals: Improve clinical efficiency and patient care, reduce duplicative processes, increase billing and coding efficiency, ensure timely access to accurate documentation.

Solution: As part of an ongoing operations integration project across five sites, the Varian Professional Services worked with TriHealth through the Varian Executive Partnership Program to improve interoperability between ARIA and Epic.

#### Results:

- Reduced duplication and improved access to accurate clinical documentation
- Identified inpatient hospitalizations, alerting admissions and clinical staff
- Improved appointment scheduling and reminder alerts to patients
- Fine-tuned billing practices, reducing missing charges and manual coding
- Increased ROI for OIS, with use at 6% pre integration to 75% post integration
- Uptick in use of best practices such as standardization, safety and timeliness
- Unprecedented access to analytics and data reporting





### Safety Statement

Side effects are related to the treatments delivered independent of our software and must be discussed by the clinician with the patient on a case-by-case basis. Medical providers retain the authority to direct all medical decisions regarding the care and treatment of its patients. Varian Medical Systems does not engage in activities that may be construed or deemed to constitute the practice of medicine without authorization. For more information, visit [www.varian.com/safety](http://www.varian.com/safety).

The testimonials contained herein are the genuine experience of the attributed individuals and may not necessarily represent the views of Varian. Individuals were/were not compensated for their testimonials.

# varian

[varian.com](http://varian.com)

#### USA, Corporate Headquarters and Manufacturer

Varian Medical Systems  
Palo Alto, CA  
Tel: 650.424.5700  
800.544.4636

#### Authorized Representative in the EU

Varian Medical Systems  
Nederland B.V.  
Kokermolen 2  
3994 DH Houten,  
The Netherlands  
[customer.relations@varian.com](mailto:customer.relations@varian.com)

#### Headquarters Europe, Eastern Europe, Middle & Near East, India, Africa

Varian Medical Systems  
International AG  
Steinhausen, Switzerland  
Tel: 41.41.749.8844

#### Asia Pacific Headquarters

Varian Medical Systems  
Pacific, Inc.  
Kowloon, Hong Kong  
Tel: 852.2724.2836

#### Australasian Headquarters

Varian Medical Systems  
Australasia Pty Ltd.  
Sydney, Australia  
Tel: 61.2.9485.0100

#### Latin American Headquarters

Varian Medical Systems  
Brasil Ltda.  
São Paulo, Brasil  
Tel: 55.11.3457.2655

Varian Medical Systems as a medical device manufacturer cannot and does not recommend specific treatment approaches. Specifications subject to change without notice. Not all features or products are available in all markets and are subject to change.

© 2020 Varian Medical Systems, Inc. All rights reserved. Varian, Varian Medical Systems, ARIA and TrueBeam are registered trademarks of Varian Medical Systems, Inc. Eclipse is a trademark of Varian Medical Systems, Inc. The names of other companies and products mentioned herein are used for identification purposes only and may be trademarks or registered trademarks of their respective owners.